TT-C-621B
May 31, 1973
SUPERSEDING
Int. Fed. Spec. TT-C-00621a(POD-ORE)
March 31, 1966 and
Fed. Spec. TT-C-621
April 17, 1942

FEDERAL SPECIFICATION

CORRECTION FLUID, DUPLICATING MACHINE STENCIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of stencil correction fluid for use on dry-process stencil paper.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).
Fed. Std. No. 313 - Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

• U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/4240

FSC 7510

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ASTM-D 1545 - Viscosity of Transparent Liquids by Bubble Time Method.
ASTM-D 1310 - Flash Point of Volatile Flammable Materials by Tag Open-Cup Apparatus.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc. Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General. The stencil correction fluid shall be a low viscosity fluid consisting of resins dissolved in a volatile solvent. The fluid shall be furnished packaged in 1-ounce glass bottles, as specified in 3.8.

3.2 Flammability.

3.2.1 Correction fluid shall have a flash point under 100°F. when tested as specified in 4.4.7. The bottle and individual packages shall be clearly labeled "Flammable" and the legend "Not Mailable Unless Packaged in Accordance with Part 125 Postal Manual" shall appear on each box unless the packaging complies with 5.3.

3.3 Viscosity. The viscosity shall be not more than tube A at 25°C., (77°F.) and shall be not less than tube A-3 at 25°C., (77°F.) when measured as specified in 4.4.1. No change in viscosity shall result when placed in storage up to 6 months in a sealed bottle furnished in accordance with this specification.

3.4 Drying properties. The fluid shall be capable of drying without any tackiness or chalking of the film when applied to dry process stencil paper and shall meet the drying test specified in 4.4.2.

3.5 Film properties. The dried film shall have good flexing properties and shall be capable of being shaved from a glass tube as specified in 4.4.4 without any crumbling of the film.

3.6 Thinning properties. The fluid shall be capable of being thinned, using the solvents specified in 4.4.3, without any precipitation or separation of components.

3.7 Performance.

3.7.1 Spreading qualities. The fluid shall be capable of being spread easily by brush in one application over a typewritten letter on dry process stencil paper so as to permit easy correction as tested in 4.4.5.

3.7.2 Working qualities. The fluid shall be capable of being easily applied over typewritten and drawn areas on dry stencil paper and shall permit correction of the treated area without any degradation in clarity or legibility of the typewritten and drawn matter on stenciled copies when tested as in 4.4.6.

3.8 Container.

3.8.1 Glass bottle. The correction fluid shall be furnished in glass bottles of one fluid ounce capacity. Each bottle shall have an opening not more than 7/16-inch in diameter and may have a suitable polyethylene adapter inserted in the neck of the bottle. The adapter shall be conical in shape with a smaller opening at the bottom than at the top, to permit the brush to be extracted from the bottle without excess fluid on it. The bottom diameter shall be the widest dimension of the bottle and shall be not less than 1-3/8 inches. The neck shall be threaded to permit the use of a screw cap cover. The glass bottle shall be of sufficient thickness to prevent breakage when a filled bottle in the individual container specified in 5.3 is dropped 6 feet. A brush or applicator securely attached to the cap and of sufficient length to reach the fluid in the bottom of the bottle shall be furnished.

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3.8.2 Bottle cap. The screw cap cover shall be polyolefin or shall have a polyolefin liner which will not affect or be affected by the fluid. The cap shall provide a tight seal without application of abnormal pressure and shall protect against any leakage.

3.9 Workmanship. The materials used in the fluid, bottle, cap, brush, adapter, and applicator shall be new, suitable for the purpose used, and free of all defects which may affect the serviceability of the finished product. Workmanship shall be in accordance with standard commercial practice.

3.10 Labeling. Each glass bottle container shall be labeled in accordance with the requirements of appropriate Federal statutes and regulations. In addition, the following information shall appear on the label in clearly discernible and legible letters:

"Correction Fluid for Dry Duplicating Stencils"

Danger - Extremely flammable
Keep away from heat, sparks or flames.
Keep out of reach of children."

The name of the manufacturer and the quantity of fluid contained in the bottle shall also appear on the label. The date of manufacture shall appear on the label or prominently marked elsewhere on the cap or paperboard box.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The contractor shall provide Government inspectors and representatives access to his plant and to all operations concerned with the manufacture and packaging of the correction compound.

4.2 Sampling.

4.2.1 For acceptance examination. The correction fluid offered for delivery shall be sampled for acceptance examination as set forth in MIL-STD-105, inspection level II. An AQL of 1.5 shall be used for major defects. An AQL of 6.5 shall be used for minor defects. Sampling shall be made from a lot consisting of all one ounce bottles of correction fluid offered for delivery at one time, or 1,000 one ounce bottles of correction fluid, which ever is the smaller.

4.2.2 Product testing. Twelve bottles of the correction fluid shall be selected at random from each batch or lot offered for delivery at one time. The tests outlined in 4.4.1 through 4.4.7 shall be performed. Failure to meet any one or more of these tests shall be cause for rejection of the entire batch or lot offered for delivery.

4.3 Inspection.

4.3.1 Acceptance examination. Each one ounce bottle of correction fluid selected in accordance with 4.2.1 shall be visually examined for conformance with this specification, and for flaws and defects in cap, bottle, brush, adapter, and applicator, adequacy of seal, leakage of fluid through cap, proper labeling and quality of workmanship. A minor defect is considered one which will not affect the useability of the bottle, cap, brush, adapter, applicator, and fluid such as superficial surface imperfections in bottle or cap; slight variations in dimensions of less than 1/64 inch, minute and infrequent bubble flaws in glass. Major defects are those which will be likely to result in failure of glass, cap, liner, adapter, detachment of applicator, or use of the compound, such as cracks in glass and cap, detachment of applicator, loss of brush bristles, (when applicable) any leakage of compound through cap, excessive out of roundness of bottle, large bubble flaws in glass, blurred or unreadable lettering on label. The sealed bottle shall be turned upside down and shaken vigorously for at least one minute to determine leakage. The cap shall then be removed and the applicator examined, dipped in the fluid and tried out with several brush strokes on stencil paper using the spreading test specified in 4.4.5. Results must be equal to that obtained with the original batch sample tested in 4.2.2.

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4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.
Content (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.4 Product testing.

4.4.1 Viscosity. Viscosity tests shall be performed with procedures specified in ASTM-D 1545. Standard viscosity reference tubes marked A to A3 should be used. If the viscosity is greater than A or less than A3, or the equivalent, in stokes or bubble seconds, the fluid shall be rejected. Several trials may be taken, if necessary, to assure precise viscosity readings.

4.4.2 Drying properties. Dip the bottom of a glass test tube (75 by 10 mm, size) to a depth of 1 inch into the correction-fluid. Remove and suspend the tube in a vertical position. Examine the upper-half of the film at the end of 45 seconds. The film shall have dried without tackiness or chalking. The test shall be carried out under conditions of 20° to 25°C, and a relative humidity of 50 to 65 percent.

4.4.3 Thinning properties. Pour 20 ml. of the correction-fluid into a test tube and allow test tube to stand unstoppered at 20° to 25°C. until the correction-fluid shall have lost by evaporation one-quarter (5 ml.) of the original volume. Add 5 ml. of a mixture of 85 percent, by volume, ethyl alcohol (95 percent), and 15 percent, by volume, ethyl ether to the concentrate, with stirring. The correction-fluid shall be in a smooth thinned condition without any precipitation or separation of components. (Note - The alcohol-ether mixture, as used herein, is not intended as a thinner in use.)

4.4.4 Film properties. Dip the bottom of a glass test tube into the correction-fluid to a depth of 1 inch. Remove and suspend the tube for 2 hours. The film shall then be capable of being shaved from the glass tube, using a razor blade. The shavings shall have good flexible properties without crumbling.

4.4.5 Spreading qualities. Over a single letter of a typewritten word on stencil paper, spread the correction-fluid with the brush that is provided. The fluid shall spread easily and sufficiently in one application. Allow to dry for 30 seconds. Type onto the treated space the desired letter. Examine: The typed letter shall appear satisfactory, with the letters adjacent to the corrected spaces unaffected.

4.4.6 Working qualities. Use dry process stencil paper and prepare as follows: Type 2 lines of capital "B" using single spacing between lines, and type three lines of the word "stencil" using a single space between words and single spacing between lines. Draw 10 parallel horizontal lines 4 inches long spaced 1/4-inch apart. Draw over the horizontal lines at an angel of 60° with the horizontal, lines spaced 1/8-inches apart (lines shall be drawn with a medium loop stylus). Treat 1/4 of the typewritten and drawn matter with the correction fluid portioning out the test area over the entire stencil. Retype, substituting "J" for "B" and "uniform" for "stencil". Remove the cushion sheet (the cushion sheet shall be capable of being suitably separated from the stencil). Place the stencil in a duplicating machine and prepared 5,000 copies. Examine several of the final copies. The corrected areas shall be equal in clarity and legibility to the remainder of the typewritten and drawn matter. Remove stencil from the duplicating machine and replace the stencil on the machine after 10 minutes. Prepare 25 copies. Examine the last copy. The corrected areas shall be equal in clarity and legibility to the typewritten and drawn matter.

4.4.7 Flash point. The flash point shall be determined according to ASTM Designation D 1310.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each filled 1-ounce bottle of fluid shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Closure of individual boxes shall be in accordance with the applicable box specification or appendix thereto.

5.1.1.2 Intermediate packaging. Twelve (12) unit packages of fluid of like description shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or to PPP-B-636. Closure of individual boxes shall be in accordance with the applicable box specification.

5.1.2 Level B. (See 5.1.1.1 and 5.1.1.2.)

5.1.3 Level C. The fluid shall be packaged to insure protection during shipment and safe delivery at destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 bottles) of fluid packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, and reinforced in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve intermediate packages (144 bottles) of fluid packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The fluid packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

5.3 Packaging for mailing. When specified (see 6.2), each filled 1 ounce bottle shall be wrapped with absorbent cushioning material to meet the requirements of the Postal Service Manual paragraph 124.22 and 124.23 part 124 and packaged as specified in 5.1.1.1.

5.4 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123 and Fed. Std. No. 313.

5.4.2 Military agencies. All marking shall be in accordance with MIL-STD-129 and Fed. Std. No. 313.

6. NOTES

6.1 Intended use. Correction fluid is to be used as specified in 1.1. Correction fluid should not be sent through the mails unless packaging, packing, and marking conform to 5.3 and 5.4. Fluid is for general use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(c) When packaging for mailing is required (see 5.3 and 6.1).

6.3 Mailing. Correction fluid should not be sent through the mails unless packaging, packing, and marking conforms to 5.3 and 5.4.

DOD Coordination has been waived.

User Interest

Army - GL

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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